Xiaochuan Li

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Research focus

I study language and cognition, especially how the structure of language shapes learning and thought. I turn questions from linguistics and cognitive science into measurements on language models, where the input can be controlled and the internals can be changed.

Education

Oct. 2025 – exp. 2027	**University of Technology Nuremberg**, Germany *M.Sc. Human and Artificial Intelligence* Coursework: deep learning, interpretability of language models, psychology and empirical study design, cognitive neuroscience.
Oct. 2024 – Jul. 2025	**University of Marburg**, Germany *German for graduate study; DSH-2 (C1 level)*
Sep. 2020 – Jun. 2024	**Shenzhen University**, China *B.A. Philosophy* Coursework: formal and modal logic, philosophy of language.

Research experience

Oct. 2026 – present	**The direction of disagreement: what departures from the majority term teach a text-only learner** *Independent project* Turned vagueness into a learning experiment: models read only naming sentences, with speakers' disagreement kept or randomly replaced, and are tested on whether they recover the order of categories. *Methods:* formal proof, transformers and LSTMs trained from scratch, fine-tuning, World Color Survey.
Sep. 2026 – present	**Modal language development in preschoolers** *M.Sc. thesis, Max Planck Institute for Human Cognitive and Brain Sciences* Behavioural study of how preschoolers acquire modal words such as *might* and *must*. In design.
Apr. – Nov. 2026	**Moral framing and dilemma choices in language models** *Project with the German Research Center for Artificial Intelligence (DFKI)* Led the utilitarian-framing branch: turned moral decisions into minimal pairs to measure how framing shifts language-model choices. Manuscript in preparation. *Methods:* probing, activation patching, low-rank analysis, steering.
Feb. 2026	**How the first information source shapes verification: a between-subjects experiment** *Small-scale psychology experiment* *Methods:* between-subjects design, Pavlovia (49 participants, 47 analysed), JASP.

May 2024	**Leibniz's logical system and characteristica universalis** *B.A. thesis, Shenzhen University* Leibniz's logic and his proposal for a universal symbolic language in which reasoning becomes calculation.

Preprints

Li, X. (2026). *The direction of disagreement: What departures from the majority term teach a text-only learner* [Preprint]. LingBuzz, lingbuzz/010504. https://lingbuzz.net/lingbuzz/010504

Research infrastructure

StemmaLM
Long-term infrastructure for language-model lineages
Turned questions about language development and transmission into model lineages raised on child-scale corpora and controlled variants: 60+ models, up to 98M parameters, public registry.
Methods: training from scratch, minimal-pair evaluation, Slurm on an HPC GPU cluster.
stemmalm.github.io

Employment

Apr. 2026 – present	**University of Technology Nuremberg**, Germany *IT Service Desk (part-time)* Student IT support; basic HPC user administration (resource allocation, access rights).
Sep. – Dec. 2021	**Shenzhen Learning Disorders Association**, China *Internship · Learning assessment at Yucai No. 2 Primary School* Ran reading assessments with primary-school children referred for learning difficulties.

Skills and languages

Human data	Experimental design, PsychoPy/Pavlovia, JASP
Evaluation	Controlled behavioural evaluation, minimal pairs
Training	Small transformers and LSTMs trained from scratch; fine-tuning
Interpretability	Activation patching, low-rank analysis, probing, steering
Tools	Python (PyTorch, Hugging Face Transformers, pandas, scikit-learn), Git, LaTeX, Slurm on an HPC GPU cluster
Languages	Mandarin Chinese (native), German (C1), English (C1), Japanese (learning)